C

LINEAR ALGEBRA • UNIT 6.01

Eigenvalues, Eigenvectors & Diagonalization

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 C	Q9 A	Q10 C
Q11 B	Q12 A	Q13 C	Q14 C	Q15 A
Q16 D	Q17 C	Q18 C	Q19 D	Q20 C
Q21 D	Q22 C	Q23 A	Q24 B	Q25 A

Eigenvalues, Eigenvectors & Diagonalization Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C 15 ✓

D 225

WORKED SOLUTION & VERIFICATION

$\|\mathbf{u}\| = \sqrt{(-9)^2 + (-12)^2} = \sqrt{225} = 15.$

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C 29 ✓

D 41

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 29$.

3 MCQ

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

- **A** $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- **C** $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
- **D** $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

- **A** (7, 6) ✓
- **B** (6, 7)
- **C** (8, 5)
- **D** (5, 8)

WORKED SOLUTION & VERIFICATION

The transformation is defined by the matrix multiplication $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$. We are given the point $(3, 1)$, so $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Performing the matrix multiplication:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (1 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 6 + 1 \\ 3 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

- **A** 1
- **B** 0 ✓
- **C** -7
- **D** 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

We are given the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2-2(1) & -1-2(2) & 0-2(5) \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right)$$

Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right)$$

This matrix is now in row-echelon form. The corresponding system of equations is:

$x + 2y = 5$

$y = 2$

Substitute the value of y into the first equation:

$x + 2(2) = 5$

$x + 4 = 5$

$x = 1$

Therefore, the unique solution is $(1, 2)$, and the value of x is 1.

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

- **A** 13 ✓
- **B** 17
- **C** 7
- **D** 169

WORKED SOLUTION & VERIFICATION

The magnitude of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$, we have $x = 5$ and $y = -12$. Therefore, $\|\mathbf{v}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C **The distance of the point (3,4) from the origin.** ✓

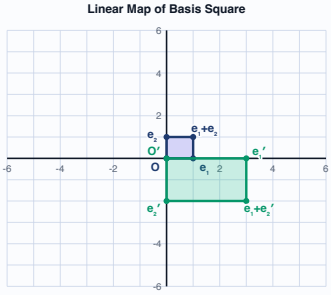
D The sum of the components of the vector.

WORKED SOLUTION & VERIFICATION

The length of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by its magnitude, $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 MCQ

Consider a linear transformation represented by the diagonal matrix $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is an eigenvector of A, what is the corresponding eigenvalue?



A 3 ✓

B -2

C 1

D 5

WORKED SOLUTION & VERIFICATION

Multiplying A by $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ gives:

$$A\mathbf{v} = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} = 3\mathbf{v}.$$

Thus, the corresponding eigenvalue is $\lambda = 3$.

10 MCQ

Consider the upper-triangular matrix $T = \begin{pmatrix} 7 & 21 \\ 0 & 4 \end{pmatrix}$. Determine the eigenvalue product $\det(T) = \lambda_1 \lambda_2$.

- A 31
- B 7
- C **28** ✓
- D 35

WORKED SOLUTION & VERIFICATION

The eigenvalues are 7 and 4, with product 28.

11 MCQ

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. If the trace of A is 7, find the other eigenvalue of A.

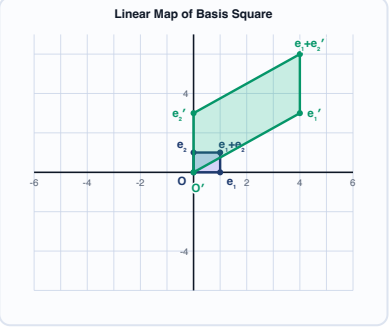
- A 1
- B **4** ✓
- C 10
- D 21

WORKED SOLUTION & VERIFICATION

Let the matrix A be $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The eigenvalues of A are the roots of the characteristic equation $\det(A - \lambda I) = 0$. We are given that $\lambda_1 = 3$ is an eigenvalue. The trace of a matrix is the sum of its diagonal elements, so $\text{tr}(A) = a + d$. We are given that $\text{tr}(A) = 7$. The trace of a matrix is also equal to the sum of its eigenvalues. Let the eigenvalues be λ_1 and λ_2. Therefore, $\lambda_1 + \lambda_2 = \text{tr}(A)$. Substituting the given values, we have $3 + \lambda_2 = 7$. Solving for λ_2, we get $\lambda_2 = 7 - 3 = 4$. The other eigenvalue of A is 4. The eigenvector information is not directly needed to find the other eigenvalue using the trace property, but it would be used if we were asked to find the matrix A itself or the determinant.

12 MCQ

Find the eigenvalues of $\begin{bmatrix} 4 & 0 \\ 3 & 3 \end{bmatrix}$.



- A $3, 4$ ✓
- B $0, 4$
- C $4, 5$
- D $-3, -4$

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $3, 4$.

13 MCQ

A 2×2 matrix has eigenvalues $-1, -5$. Find its trace.

- A 6
- B 5
- C -6 ✓
- D -5

WORKED SOLUTION & VERIFICATION

The trace is the sum of the eigenvalues (with multiplicity): -6.

14 MCQ

Find the eigenvalues of $\begin{bmatrix} 4 & 0 & 0 \\ -2 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

- A $0, 4$
- B $1, 2, 5$
- C $0, 1, 4$ ✓
- D $0, -1, -4$

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $0, 1, 4$.

15 MCQ

A 2×2 matrix has eigenvalues $5, 0$. Find its trace.

- A 5 ✓
- B 0
- C 6
- D -5

WORKED SOLUTION & VERIFICATION

The trace is the sum of the eigenvalues (with multiplicity): 5.

16 MCQ

Find the eigenvalues of $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 5 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

- A $-2, -1$
- B $0, 1, 6$
- C $1, 0, -5$
- D $-1, 0, 5$ ✓

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $-1, 0, 5$.

17 MCQ

A 2×2 matrix has eigenvalues $6, -5$. Find its trace.

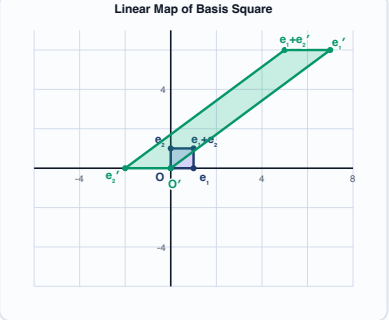
A -30	B 2
C 1 ✓	D -1

WORKED SOLUTION & VERIFICATION

The trace is the sum of the eigenvalues (with multiplicity): 1.

18 MCQ

$\mathbf{v} = \langle 4, 6 \rangle$ is an eigenvector of $A = \begin{bmatrix} 7 & -2 \\ 6 & 0 \end{bmatrix}$. What is its eigenvalue?



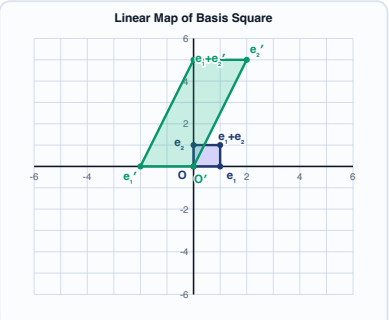
A 3	B 5
C 4 ✓	D -4

WORKED SOLUTION & VERIFICATION

$A\mathbf{v} = \langle 16, 24 \rangle = 4\langle 4, 6 \rangle$, so $\lambda = 4$.

19 MCQ

Find the eigenvalues of $\begin{bmatrix} -2 & 2 \\ 0 & 5 \end{bmatrix}$.



A $2, -5$	B $-2, 2$
C $-1, 6$	D $-2, 5$ ✓

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $-2, 5$.

20 MCQ

A 3×3 matrix has eigenvalues $-5, 2, 2$. Find its determinant.

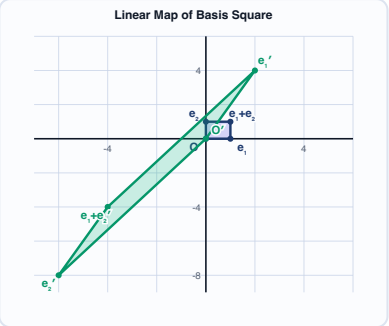
- **A** -1
- **B** -19
- **C** -20 ✓
- **D** 20

WORKED SOLUTION & VERIFICATION

The determinant is the product of the eigenvalues (with multiplicity): -20.

21 MCQ

$\mathbf{v} = \langle -6, -4 \rangle$ is an eigenvector of $A = \begin{bmatrix} 2 & -6 \\ 4 & -8 \end{bmatrix}$. What is its eigenvalue?



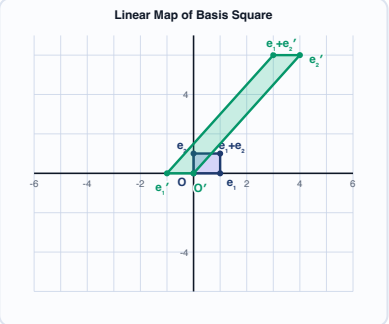
- **A** -4
- **B** -1
- **C** 2
- **D** -2 ✓

WORKED SOLUTION & VERIFICATION

$A\mathbf{v} = \langle 12, 8 \rangle = -2\langle -6, -4 \rangle$, so $\lambda = -2$.

22 MCQ

Calculate the eigenvalues of $\begin{bmatrix} -1 & 4 \\ 0 & 6 \end{bmatrix}$.



- **A** $1, -6$
- **B** $-1, 4$
- **C** $-1, 6$ ✓
- **D** $0, 7$

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $-1, 6$.

23 MCQ

A 2×2 matrix has eigenvalues $6, -4$. Find its trace.

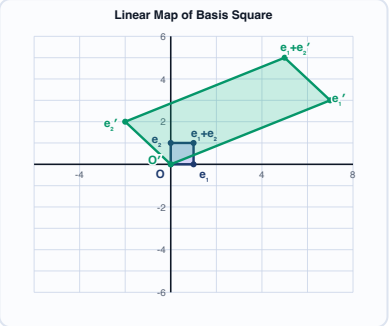
- **A** 2 ✓
- **B** 3
- **C** -2
- **D** -24

WORKED SOLUTION & VERIFICATION

The trace is the sum of the eigenvalues (with multiplicity): 2.

24 MCQ

$\mathbf{v} = \langle -4, -6 \rangle$ is an eigenvector of $A = \begin{bmatrix} 7 & -2 \\ 3 & 2 \end{bmatrix}$. What is its eigenvalue?



- **A** 5
- **B** 4 ✓
- **C** -4
- **D** -40

WORKED SOLUTION & VERIFICATION

$A\mathbf{v} = \langle -16, -24 \rangle = 4\langle -4, -6 \rangle$, so $\lambda = 4$.

25 MCQ

Find the eigenvalues of $\begin{bmatrix} 6 & 0 & 0 \\ -1 & -4 & 0 \\ -3 & 2 & 6 \end{bmatrix}$.

- **A** $-4, 6$ ✓
- **B** $0, 6$
- **C** $-3, 7$
- **D** $4, -6$

WORKED SOLUTION & VERIFICATION

The matrix is triangular, so its eigenvalues are its diagonal entries: $-4, 6$.